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ORIGINAL RESEARCH

Diagnostic Accuracy of Cardiac Computed Tomography and 18-F Fluorodeoxyglucose Positron Emission Tomography in Cardiac Masses

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ABSTRACT

OBJECTIVES This study sought to assess the diagnostic accuracy of cardiac computed tomography (CT) and ^{18}F-fluorodeoxyglucose (^{18}F-FDG) with positron emission tomography/computed tomography (PET/CT) in defining the nature of cardiac masses.

BACKGROUND The diagnostic accuracy of cardiac CT and ^{18}F-FDG PET/CT in identifying the nature of cardiac masses has been analyzed to date only in small samples.

METHODS Of 223 patients with echocardiographically diagnosed cardiac masses, a cohort of 60 cases who underwent cardiac CT and ^{18}F-FDG PET/CT was selected. All masses had histological confirmation, except for a minority of thrombotic formations. For each mass, 8 morphological CT signs, standardized uptake value (SUV_{max}, SUV_{mean}), metabolic tumor volume, and total lesion glycolysis in ^{18}F-FDG PET were used as diagnostic markers.

RESULTS Irregular tumor margins, pericardial effusion, invasion, solid nature, mass diameter, CT contrast uptake, and pre-contrast characteristics were strongly associated with the malignant nature of masses. The coexistence of at least 5 CT signs perfectly identified malignant masses, whereas the detection of 3 or 4 CT signs did not accurately discriminate the masses' nature. The mean SUV_{max}, SUV_{mean}, metabolic tumor volume, and total lesion glycolysis values were significantly higher in malignant than in benign masses. The diagnostic accuracy of SUV, metabolic tumor volume, and total lesion glycolysis ^{18}F-FDG PET/CT parameters was excellent in detecting malignant masses. Among patients with 3 or 4 pathological CT signs, the presence of at least 1 abnormal ^{18}F-FDG PET/CT parameter significantly increased the identification of malignancies.

CONCLUSIONS Cardiac CT is a powerful tool to diagnose cardiac masses as the number of abnormal signs was found to correlate with the lesions' nature. Similarly, ^{18}F-FDG PET/CT accurately identified malignant masses and contributed with additional valuable information in diagnostic uncertainties after cardiac CT. These imaging tools should be performed in specific clinical settings such as involvement of great vessels or for disease-staging purposes.
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Cardiac masses are a diagnostic dilemma for clinicians as the discrimination between benign and malignant lesions is often challenging. Overall, cardiac masses can be classified into 4 subgroups: primary cardiac benign tumors; primary malignant tumors; secondary malignant tumors; pseudotumors (1). Primary cardiac tumors are rare and approximately 20% to 25% are malignant; secondary cardiac tumors metastasizing into the heart occur nearly 20× more frequently than primary cardiac masses do (2). Pseudotumors are common in cardiovascular clinical practice and encompass several clinical conditions, ranging from "normal" anatomic and congenital variants to other formations such as thrombi (3).

An appropriate assessment the extent of masses is essential to address therapeutic strategies, varying from medical management to aggressive surgical intervention (4). Most importantly, the nature of cardiac masses as well as the subsequent treatment can affect prognosis; in fact, the majority of cardiac benign masses have an excellent outcome (5), whereas patients with primary or secondary malignant cardiac masses show a poor prognosis (6-8).

An integrated noninvasive multimodality imaging approach is helpful to evaluate and diagnose cardiac masses as catheter-based biopsy is often unfeasible or may be inconclusive (4). Echocardiography remains the first-line imaging modality for a suspected cardiac mass because of its availability, noninvasiveness, absence of contrast material or radiation exposure, and its ability to provide a dynamic assessment (9). However, it is unable to allow a global evaluation of cardiac and extracardiac structures as well as a tissue characterization. Cardiac computed tomography (CT) and cardiac magnetic resonance (CMR) may confer valuable additional information, regarding anatomy and tissue substrate (10). Nevertheless, these noninvasive imaging techniques do not always differentiate benign from malignant cardiac masses. On the other hand, molecular imaging methods such as ^{18}F-fluorodeoxyglucose (^{18}F-FDG) with positron emission tomography (PET) can visualize cell metabolism and thereby assess metabolic activity (11). Published data regarding cardiac CT findings and metabolic activity of cardiac masses are scarce and inconclusive.

We therefore sought to test the hypothesis that cardiac CT findings and ^{18}F-FDG PET/CT signals may increase the diagnostic accuracy in cardiac masses.

ABBREVIATIONS AND ACRONYMS

^{18}F-FDG = ^{18}F-fluorodeoxyglucose

AUC = area under the curve

CI = confidence interval

CT = computed tomography

CMR = cardiac magnetic resonance

MTV = metabolic tumor volume

NPV = negative predictive value

PET = positron emission tomography

PPV = positive predictive value

ROC = receiver-operating characteristic

SUV = standardized uptake value

TLG = total lesion glycolysis

METHODS

STUDY DESIGN. We retrospectively evaluated all patients who underwent imaging techniques for suspected cardiac masses between January 1, 2000, and December 31, 2016. Of the 178 patients identified from our echocardiographic database, we enrolled those who underwent a cardiac CT and ^{18}F-FDG PET/CT within 1 month from diagnosis.

From January 1, 2017, to December 31, 2018, we prospectively enrolled all consecutive patients admitted to our institution for suspected cardiac masses who underwent both cardiac CT and ^{18}F-FDG PET/CT (**Figure 1**). After the diagnostic process, patients were classified as having benign masses or malignant masses and subsequently into 4 subtypes: pseudotumors; primary cardiac benign tumors; primary malignant cardiac tumors; or secondary cardiac tumors.

All-cause mortality was ascertained throughout the review of medical records.

All patients were treated in accordance with the Declaration of Helsinki and provided informed consent for anonymous publication of scientific data. The study protocol received approval from the local ethics committee (Comitato Etico Indipendente dell'Azienda Ospedaliero-Universitaria di Bologna, Policlinico S. Orsola-Malpighi, registered number102/2017/0/Oss).

DEFINITIONS AND PATHOLOGY. Benign and malignant primary cardiac tumors were classified according to the World Health Organization histological classification of tumors of the heart (12). We defined

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the *JACC: Cardiovascular Imaging* author instructions page.

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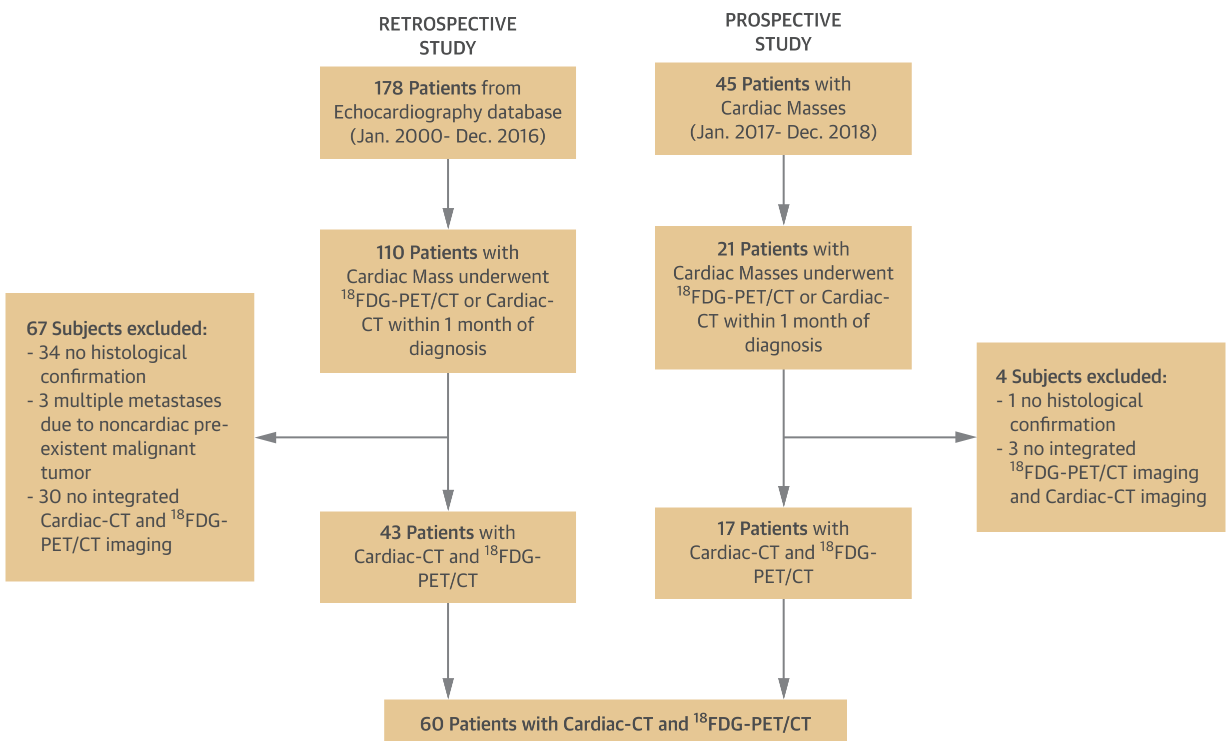
FIGURE 1 Study Flowchart



The study flowchart shows the retrospective and prospective enrollment of our population as well as the excluded patients. CT = computed tomography; ^{18}F-FDG= ^{18}F-fluorodeoxyglucose; PET = positron emission tomography.

pseudotumors as any benign process capable of producing a significantly sized mass, without tumor characteristics (13).

We excluded normal anatomic variants due to the impossibility of obtaining histological examination and patients with clinical and laboratory data of active infective endocarditis (14).

All cases were histopathologically confirmed independently by 2 experienced pathologists according to the American Society of Clinical Oncology/College of American Pathologists 2010 criteria, apart from thrombi, in which the diagnosis was confirmed by the radiological evidence of thrombus resolution after appropriate anticoagulant treatment.

Cardiac tissue masses were evaluated following surgical resection, biopsies, or autopsy.

MULTIDETECTOR CARDIAC CT. All cardiac CTs were performed using a 128-row scanner (Brilliance iCT SP, Philips, Amsterdam, the Netherlands) or a 64-row scanner (Somatom Sensation 64 Cardiac, Siemens, Forchheim, Germany), with electrocardiographic retrospective synchronization.

Two expert radiologists (each with more than 10 years of experience in the cardiac imaging), blinded to clinical information, independently evaluated the following imaging features of the cardiac masses: nature (mainly solid vs. mainly cystic) according to mass density, rated as hypo-, iso- or hyperdense as compared with the normal cardiac muscle; size; intramass calcifications; contrast uptake, defined as the increase (≥10 Hounsfield units) in cardiac mass density compared with baseline; tumor margins, described as circumscribed, microlobulated, obscured (partially hidden by adjacent tissue), indistinct (ill-defined), or spiculated (characterized by lines radiating from the mass); pericardial effusion, defined as a fluid accumulation between the 2 pericardial layers; invasiveness, defined as disruption of neighboring tissue and extension of the mass into the tissue. Disagreements in qualitative evaluations were resolved by a one-third consensus re-evaluation.

^{18}F-FDG WITH PET. To reduce the unpredictable physiological myocardial FDG uptake, all patients

TABLE 1 Histological Characterization of Benign and Malignant Masses

Benign masses (n = 20)	
Primitive benign tumors	
Mixoma	87.5 (7/8)
Paraganglioma	12.5 (1/8)
Pseudotumors	
Thrombi	33.3 (4/12)
Noninfectious vegetations	25.0 (3/12)
Lipomatosis	16.7 (2/12)
Pericardial cystis	16.7 (2/12)
Calcifications	8.3 (1/12)
Malignant masses (n = 40)	
Primitive malignant tumors	
Sarcoma	72.2 (13/18)
Lymphoma B cells	27.8 (5/18)
Metastasis	
Lymphoma	50.0 (11/22)
Melanoma	13.6 (3/22)
Mediastinal sarcoma	13.6 (3/22)
Choriocarcinoma	4.5 (1/22)
Lung cancer	4.5 (1/22)
Bladder cancer	4.5 (1/22)
Colon carcinoma	4.5 (1/22)
Hepatocellular carcinoma	4.5 (1/22)

Values are % (n/N).

were observed after at least 12 h of fasting with a last fatty meal without carbohydrates as previously described (15).

PET/CT scans were evaluated by 2 experienced nuclear medicine physicians who assessed attenuation-corrected images.

In particular, the following intensity- and volume-based PET parameters were measured using PET/CT images to draw an automatically defined region of interest on the cardiac mass: maximum standardized uptake value (SUV_{max}); mean standardized uptake value (SUV_{mean}); metabolic tumor volume (MTV), (i.e., volume of interest consisting of all spatially connected voxels within a fixed threshold of 40% of the SUV_{max}); and total lesion glycolysis (TLG) (as the product of MLV and SUV_{mean}). When the cardiac mass did not show an increased radiotracer uptake, the automatically defined region of interest was not sufficient to measure MTV and TLG, and thus only SUV_{max} and SUV_{mean} were automatically calculated. For all patients, to compare the metabolic parameters assessed from the various examinations, the blood pool SUV_{max} was collected in the upper vena cava at the height of the highest point of the aortic arch.

All parameters were collected on a continuous scale and subsequently analyzed in relation to the underlying histopathological substrate to identify appropriate cutoffs to discriminate between benign and malignant masses.

Additional information regarding pathology, cardiac CT, and ^{18}F-FDG PET acquisitions are specified in the Supplemental Appendix.

STATISTICAL ANALYSIS. First, we compared the clinical characteristics, laboratory parameters, and morphological features (assessed through echocardiography, CT, PET/CT) of patients with benign and malignant cardiac tumors. Chi-square and Mann-Whitney *U* tests were used for categorical and continuous variables, respectively. The lesions were further characterized as primary benign, primary malignant, secondary (metastases), pseudotumors, and the above-mentioned variables were compared among the 4 groups using chi-square and Kruskal-Wallis tests. If the overall test was significant, post hoc pairwise comparisons were performed at the Bonferroni-corrected significance level of 0.0083 to control for type I error.

Second, we evaluated the diagnostic accuracy of PET and CT signs in predicting a diagnosis of malignant cardiac lesion, either primary or secondary, and for each sign (irregular tumor margin; pericardial effusion; presence of calcification; contrast enhancement, invasion, dimension, density, pre-contrast characteristics) we computed the sensitivity,

TABLE 2 Comparison of Clinical and Laboratory Parameters of Cardiac Benign and Malignant Masses

	Benign Cardiac Masses (n = 20)	Malignant Cardiac Masses (n = 40)	p Value
Age, yrs	60.4 ± 17	58.4 ± 16.1	0.70
Male	10 (50.0)	27 (67.5)	0.20
BMI, kg/m^2	25.3 ± 4.9	26.6 ± 5	0.30
Current cigarette smoking	9 (45)	16 (40)	0.70
Hypertension	14 (70)	20 (50)	0.14
Hypercholesterolemia	8 (40.0)	13 (32.5)	0.60
Diabetes mellitus	3 (15)	4 (10)	0.60
Family history of cancer	0 (0)	9 (22)	0.03
Previous cancer	9 (45.0)	15 (37.5)	0.60
Incidental diagnosis	9 (45)	6 (15)	0.011
Localization			<0.001
Right side	17 (42.5)	5 (25.0)	
Left side	3 (7.5)	12 (60.0)	
Pericardium and great vessels	20 (50)	3 (15)	
Laboratory parameters			
GFR, ml/min per 1.73 m^2	81.6 ± 24.8	85.0 ± 26.2	0.60
Serum creatinine levels, mg/dl	1 ± 0.8	1 ± 0.4	0.60
Leukocyte count, n/mm^3	8,624 ± 3,760	8,072 ± 4,336	0.60
Serum CRP levels, mg/dl	2.0 ± 3.2	3.5 ± 4.3	0.30

Values are mean ± SD or n (%).
BMI = body mass index; CRP = C-reactive protein; GFR = glomerular filtration rate.

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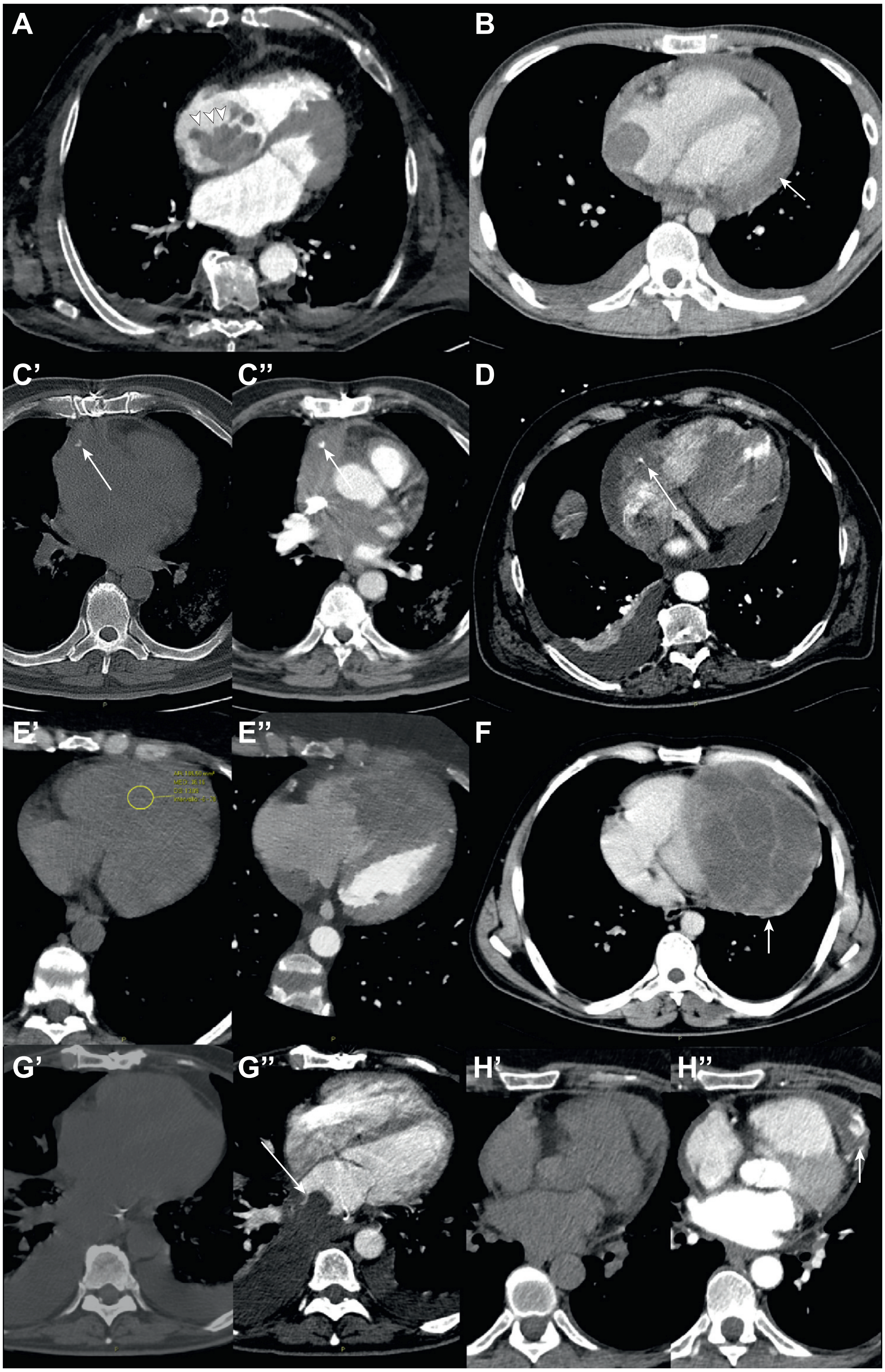
FIGURE 2 CT Images of Benign and Malignant Cardiac Masses



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specificity, positive predictive value (PPV) and negative predictive value (NPV), and the overall accuracy, which is the percentage of patients correctly classified. Ninety-five percent confidence intervals (CIs) for proportions were calculated according to the Newcombe efficient-score method (16).

Third, we estimated the diagnostic accuracy of glucose uptake (quantified by ^{18}F-FDG PET SUV), MTV, TLG, and of the number of signs in discriminating malignant from benign lesions using receiver-operating characteristic (ROC) curve analysis.

Lastly, we analyzed the mortality of the study cohort and the predictive ability of glucose uptake and of the number of CT signs using Kaplan-Meier survival estimates and Cox regression analysis.

The significance level was set to $p < 0.05$, and all analyses were performed using Stata version 13.1 (Stata Corp., College Station, Texas) and IBM SPSS version 25.0 (IBM Corp., Armonk, New York).

RESULTS

Of 223 patients with an echocardiographic suspicion of cardiac masses, 131 underwent cardiac CT or ^{18}F-FDG PET/CT. Thirty-three patients were excluded because they did not complete the radiological procedure, 35 due to the absence of histological examination, and 3 patients because they had extensive multiorgan metastases. The final sample therefore included 60 patients (**Figure 1**). Cardiac tissue masses were evaluated following surgical resection in 31 patients, by biopsies in 24 patients, by autopsy in 1, and with radiological resolution after adequate anticoagulant therapy for 4 thrombi. Based on the histopathologic examination, we identified 20 patients with benign cardiac masses (8 primary benign tumors and 12 pseudotumors) and 40 patients with malignant masses (18 primary malignant cardiac tumors and 22 metastases) (**Table 1**).

According to the study design and classification, the distribution of cardiac masses is shown in Supplemental Figure 1 and Supplemental Table 1.

TABLE 3 Comparison of Morphologic (PET/CT) Parameters of Cardiac Benign and Malignant Masses

	Benign Cardiac Masses (n = 20)	Malignant Cardiac Masses (n = 40)	p Value
Cardiac CT			
Irregular tumor margins	0 (0.0)	35 (87.5)	<0.001
Pericardial effusion	1 (5.0)	29 (72.5)	<0.001
Calcification	3 (15)	2 (5)	0.20
Invasion	0 (0.0)	31 (77.5)	<0.001
Solid density	17 (85)	38 (95)	0.028
Diameter >30 mm	3 (15)	30 (75)	<0.001
Pre-contrast characteristics			0.011
Isodense	13 (65.0)	37 (92.5)	
Hypodense	3 (15)	0 (0)	
Hyperdense	4 (20.0)	3 (7.5)	
CT contrast uptake	7 (35.0)	39 (97.5)	<0.001
^{18}F-FDG PET			
Noncardiac ^{18}F-FDG uptake	13 (65)	32 (80)	0.30
^{18}F-FDG SUV_{max}	2.8 ± 1.2	12.6 ± 7.7	<0.001
SUV_{max}			
Cutoff ≥4.9	5.6	86.8	<0.001
^{18}F-FDG SUV_{mean}	2 ± 0.9	6.8 ± 3	<0.001
^{18}F-FDG SUV_{BP}	1.9 ± 0.4	1.8 ± 0.5	0.35
Metabolic tumor volume, ml	2.6 (1.7–4.2)	44.7 (9.9–175.5)	0.001
Total lesion glycolysis, g	6.5 (2.0–14.6)	239.1 (69.7–1,614.1)	<0.001

Values are n (%), mean ± SD, or median (interquartile range).

BP = blood pressure; CT = computed tomography; ^{18}F-FDG = ^{18}F-fluorodeoxyglucose; SUV = standardized uptake value.

Patient characteristics are summarized in **Table 2**. Overall, mean age at diagnosis was 59.1 ± 16.3 years with no difference between male and female patients. The comparison of demographic and clinical characteristics indicated that the 2 groups (benign vs. malignant) were similar in sex distribution, mean age, risk factors, and laboratory parameters. However, patients with malignant lesions were more likely to have a family history of cancer and less likely to have an incidental diagnosis. Interestingly, benign masses were predominantly located in the left heart chambers, whereas malignant ones—both primitive and metastatic—were often right-sided and detected in the pericardium and/or great vessels.

FIGURE 2 Continued

(A to H) Illustrations of the different imaging features of benign and malignant cardiac masses evaluated on computed tomography (CT). **(A)** CT axial plane demonstrates a cardiac mass in the right atrium with irregular tumor margins **(arrowheads). (B)** CT axial plane shows a cardiac mass in the right atrium with pericardial effusion **(arrow). (C′, C″)** CT axial plane illustrates a calcification in a right cardiac mass that is well defined in the basal image **(arrow** in **C′)**, substantially unchanged on post contrast image **(arrow in C″). (D)** CT axial plane highlights a right cardiac mass that encases the right coronary artery determining the "tear drop" sign strongly suggestive for vascular invasion **(arrow). (E′, E″)** CT axial plane exhibits a cardiac mass in the right ventricle that is well identifiable after contrast media injection **(E″)** with a solid density (36 Hounsfield units in **E′**). **(F)** CT axial plane reveals a large cardiac mass with a maximum diameter >15 cm. **(G′, G″)** CT axial plane displays a cardiac mass in the left atrium that is well identifiable after contrast media injection **(arrow in G″)** showing isodensity on basal image **(G′). (H′, H″)** CT axial plane proves a cardiac mass with strong contrast uptake **(arrow in H″)** that is well identifiable as compared to the basal image **(H′).**

TABLE 4 Comparison of Imaging (CT/PET) Parameters Among the Different Types of Cardiac Masses

	Primary Benign Cardiac Tumors (n = 8)	Primary Malignant Cardiac Tumors (n = 18)	Secondary Cardiac Tumors (n = 22)	Pseudotumors (n = 12)	p Value*
Cardiac CT					
Irregular tumor margins	0 (0.0)	17 (94.4)†	18 (81.8)†	0 (0.0)	<0.001
Pericardial effusion	0 (0.0)	14 (77.8)†	15 (68.2)†	1 (8.3)	<0.001
Calcification	0 (0.0)	0 (0.0)	2 (9.1)	3 (25.0)	0.08
Invasion	0 (0.0)	12 (66.7)†	19 (86.4)†	0 (0.0)	<0.001
Solid density	8 (100.0)	18 (100.0)	20 (90.9)	9 (75.0)	0.08
Diameter >30 mm	0 (0.0)	16 (88.9)†	14 (63.6)†	3 (25.0)	<0.001
Pre-contrast characteristics					<0.001
Isodense	4 (50.0)‡	17 (94.4)	20 (90.9)	9 (75.0)	
Hypodense	0 (0.0)	0 (0.0)	0 (0.0)	3 (25.0)	
Hyperdense	4 (50.0)§	1 (5.6)	2 (9.1)	0 (0.0)	
TC contrast uptake	6 (75.0)	18 (100.0)	21 (95.5)	1 (8.3)‖	<0.001
^{18}F-FDG PET					
Noncardiac ^{18}F-FDG uptake	4 (50.0)	12 (66.7)	20 (90.9)	9 (75.0)	0.20
^{18}F-FDG SUV_{max}	2.33 ± 0.85	12.7 ± 8.81†	12.5 ± 6.76†	3.15 ± 1.36	<0.001
SUV_{max}					
Cut-off ≥4.9	0.0	83.3†	90.0†	10.0	<0.001
^{18}F-FDG SUV_{mean}	1.74 ± 0.66	6.29 ± 2.61†	7.21 ± 3.25†	2.24 ± 1.00	<0.001
^{18}F-FDG SUV_{BP}	1.96 ± 0.55	1.76 ± 0.45	1.77 ± 0.58	1.86 ± 0.27	0.80
Metabolic tumor volume, ml	–	26.7 (9.4–77.4)	105 (9.7–261.5)¶	2.6 (1.7–4.2)	0.003
Total lesion glycolysis, g	–	188 (62.6–392.1)	310 (92.7–1,687.1)¶	6.5 (2.0–14.6)	0.001

Values are n (%), mean ± SD, %, or median (interquartile range). The p values were determined using. *Chi-square test for categorical variables, analysis of variance for normally distributed variables, and Kruskal-Wallis test for non-normally distributed continuous variables. Following significant omnibus tests, post hoc pairwise comparisons were carried out at the Bonferroni-corrected significance level $p < 0.0083$. Post hoc significant comparisons include the following: †primary malignant cardiac tumors and secondary cardiac tumors > primary benign cardiac tumors and pseudotumors; ‡primary benign cardiac tumors < primary malignant cardiac tumors, secondary cardiac tumors; §primary benign cardiac tumors < primary malignant cardiac tumors, secondary cardiac tumors, and pseudotumors; ‖pseudotumors < primary malignant cardiac tumors, secondary cardiac tumors, and primary benign cardiac tumors; and ¶secondary cardiac tumors > pseudotumors.

Abbreviations as in Table 3.

Patient characteristics categorized by tumor histology are shown in Supplemental Table 2; there was a significant variation in sex difference as female patients were disproportionately affected by primary cardiac benign tumors (62.5%), whereas male patients were more represented in the metastasis group (77.3%). Cardiovascular risk factors were similar across the different tumor types; however, patients with primary benign tumors had a higher incidence of hypercholesterolemia. One-third of the patients with cardiac metastasis had a documented family history of cancer. Overall, 15 incidental masses were detected: 5 of 12 in the pseudotumor group, 5 of 22 in the metastasis group, and 50% in the primary benign tumor group; only 1 case was accidentally diagnosed in the primary malignant tumor group.

COMPARISON OF CT CHARACTERISTICS AND HISTOLOGICAL SUBTYPE. Thirty-five patients (58.3%) had irregular tumor margins, all with malignant masses (Figure 2). Specifically, 17 patients (94.4%) of the 18 primary malignant cardiac masses and 18 (81.8%) of the 22 in the metastasis group had irregular tumor margins.

Pericardial effusion was present in 30 patients (50%): only 1 had a benign mass and the other 29 had a malignant neoformation. Among patients with malignancies, pericardial effusion was more frequent in the primary malignant group (77.8%) as compared to the metastasis group (68.2%) (Figure 2).

The presence of calcifications was unrelated to the nature of masses as they were equally distributed between benign and malignant lesions.

Thirty-one patients had signs of invasion at cardiac CT, all in the malignant group. The solid nature of the mass and the size >3 cm were more frequent in the malignant group. ROC analysis indicated an excellent performance of a mass diameter >3 cm as an indicator of malignancy (area under the curve [AUC]: 0.837; 95% CI: 0.738 to 0.936).

A total of 46 masses had uptake of contrast agent, 39 in the malignant groups and 7 among benign cardiac masses. In particular, 75% of primary benign tumors and only 8.3% of pseudotumors showed contrast uptake. On the other hand, contrast uptake was equally observed in primary and secondary malignant masses (Tables 3 and 4).

TABLE 5 Diagnostic Accuracy of Selected CT Signs and 18F-FDG PET Parameters in Predicting the Malignant Nature of Cardiac Masses

	Sensitivity (95% CI)	Specificity (95%CI)	PPV (95% CI)	NPV (95% CI)	Accuracy
Irregular margins	87.5 (75.8-94.6)	100.0 (92.5-100.0)	100.0 (92.5-100.0)	80.0 (67.2-89.1)	91.7
Pericardial effusion	72.5 (59.2-83.0)	95.0 (85.1-99.0)	96.7 (87.4-100.0)	63.3 (49.8-75.2)	80.0
Calcification	5.0 (1.3-14.1)	85.0 (72.9-92.8)	40.0 (27.8-53.4)	30.9 (19.9-44.1)	31.7
Invasion	77.5 (64.5-87.1)	100.0 (92.5-100.0)	100.0 (92.5-100.0)	68.9 (55.5-80.0)	85.0
Size >3 cm	75.0 (61.8-85.1)	85.0 (72.9-92.8)	90.0 (79.9-96.9)	62.9 (49.5-74.9)	78.3
Solid component	95.0 (85.2-99.4)	15.0 (7.5-26.7)	69.1 (55.7-80.2)	60.0 (46.5-72.2)	68.3
Isodense signal	92.5 (81.9-97.9)	35.0 (23.4-48.3)	74.0 (60.8-84.3)	70.0 (56.6-80.9)	73.3
Contrast enhancement	97.5 (88.6-100.0)	65.0 (51.5-76.7)	84.78 (72.7-92.7)	92.8 (82.2-92.7)	86.7
^{18}F-FDG PET Parameters					
SUV_{max} ≥4.9	86.8 (75.0-94.1)	94.4 (84.4-99.1)	97.0 (88.0-100.0)	77.3 (64.3-86.9)	89.3
TLG ≥29	90.0 (78.8-96.3)	100.0 (92.5-101.4)	100.0 (92.5-101.4)	66.7 (53.2-78.1)	91.7
MTV ≥8.2	83.3 (71.0-91.6)	100.0 (92.5-101.4)	100.0 (92.5-101.4)	54.5 (41.2-67.2)	86.1

Values are %, unless otherwise indicated.
CI = confidence interval; MTV = metabolic tumor volume; NPV = negative predictive value; PPV = positive predictive value; TLG = total lesion glycolysis; other abbreviations as in Table 3.

The diagnostic accuracy of CT parameters is shown in Table 5. Invasion and irregular margins were present only among patients with malignancies, thus the PPV of these signs was 1.

The diagnostic accuracy of the other 6 signs ranged from a maximum of 86.7% for CT contrast uptake to a minimum of 31.7% for calcifications.

Using ROC analysis, we found an AUC of 0.988 and 95% CI of 0.969 to 1.000. A cutoff ≥5 CT signs perfectly predicted malignant masses (PPV = 100%), whereas a cutoff of ≤2 CT signs perfectly excluded malignancies (NPV = 100%). In the presence of 3 or 4 CT signs, the PPV was 87% and did not allow us to accurately predict the nature of cardiac masses.

COMPARISON OF CARDIAC PET CHARACTERISTICS AND HISTOLOGICAL SUBTYPE. Baseline functional PET parameters are depicted in Table 3. In 53 subjects (88.3%) an optimal suppression of myocardial ^{18}F-FDG uptake was observed.

SUV_{max}, SUV_{mean}, MTV, and TLG mean values were significantly higher in malignant cardiac masses than in benign lesions. Among the benign groups, patients with a primary tumor had lower SUV_{max}, SUV_{mean}, MTV, and TLG as compared to the pseudotumor group although the difference was not statistically significant. Similarly, in patients with malignancies, the metabolic tumor indices did not differ between primary tumors and metastasis (Table 4).

As shown in Table 5, SUV, MTV, and TLG accurately identified malignant lesions. Accordingly, ROC analysis confirmed an excellent diagnostic performance of such parameters: SUV (AUC: 0.948; 95% CI: 0.891 to 1.000); MTV (AUC: 0.928; 95% CI: 0.841 to 1.000); and TLG (AUC: 0.961; 95% CI: 0.902 to 1.000). The optimal threshold for these parameters was ≥4.9 for SUV, ≥29 for TLG, and ≥8.2 for MTV.

Figure 3 illustrates examples of cardiac CT and ^{18}F-FDG PET/CT images according to the 4 subtypes.

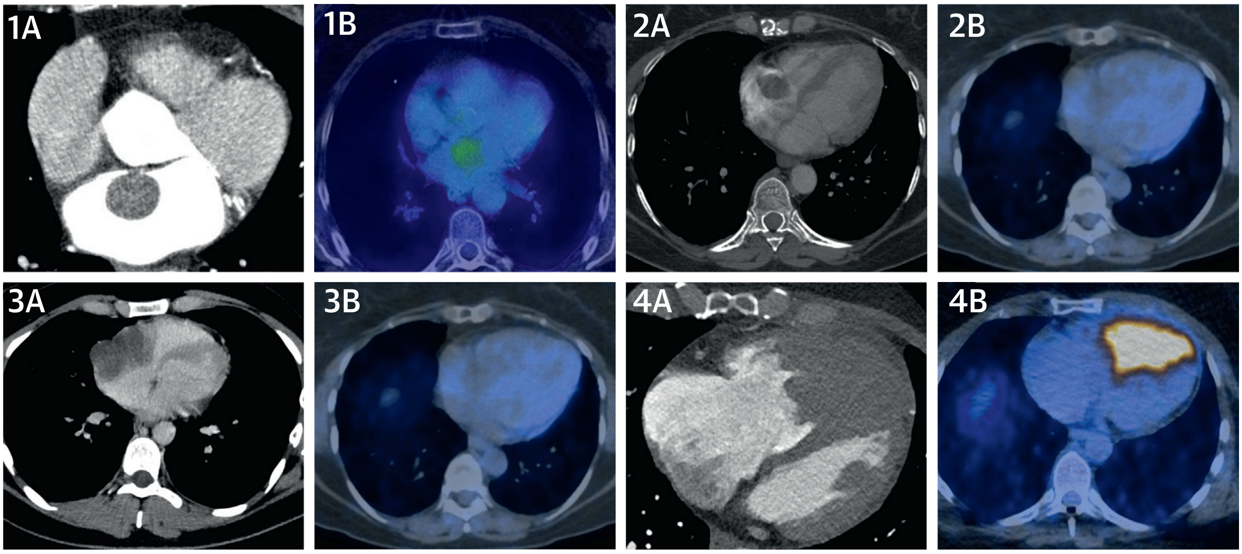
THE ROLE OF ^{18}F-FDG PET/CT IN CASE OF DIAGNOSTIC UNCERTAINTY AFTER CARDIAC CT. Among the 13 patients with a diagnostic uncertainty after cardiac CT (exhibiting 3 or 4 signs), ^{18}F-FDG PET/CT provided additional information. In fact, ^{18}F-FDG PET/CT was able to correctly identify a malignant tumor (either primitive or metastatic) in 5 of 7 cases, as at least 1 parameter (SUV, MTV, or TLG) was above the cutoff value we identified as optimal. Similarly, all 6 patients with benign masses presented ^{18}F-FDG PET/CT parameters below the cutoff values. Therefore, in uncertain cases—patients with 3 or 4 CT signs—the addition of at least 1 PET parameter increased both specificity and PPV to 100% (Central Illustration).

OUTCOME ANALYSES. Of the 59 patients (1 was lost during follow-up) followed for a median time of 18.1 months, 31 died, 30 subjects were diagnosed with malignant lesions, and 1 patient with a cardiac myxoma. In a univariate Cox regression model, the number of CT signs significantly predicted patients' mortality (hazard ratio: 1.342; 95% CI: 1.101 to 1.636). Each sign conferred a 34.2% increase in the mortality risk. We then investigated the predictive role of FDG markers in a forward stepwise multiple Cox regression model. Only SUV_{max} independently predicted the mortality risk (hazard ratio: 3.145; 95% CI: 1.341 to 7.373).

DISCUSSION

In this study we investigated the diagnostic performance of cardiac CT and ^{18}F-FDG PET/CT in 60 patients with histologically confirmed cardiac masses.

FIGURE 3 **Cardiac CT and ^{18}F-FDG PET in Cardiac Masses**



(1A) Cardiac CT and **(1B)** ^{18}F-FDG PET images of a left atrial myxoma. **(2A)** Cardiac CT and **(2B)** ^{18}F-FDG PET images of a right atrial thrombus. **(3A)** Cardiac CT and **(3B)** ^{18}F-FDG PET images of a cardiac angiosarcoma. **(4A)** Cardiac CT and **(4B)** ^{18}F-FDG PET images of a melanoma metastasis. Abbreviations as in Figure 1.

The main novelty of our work is that for the first time we assessed the diagnostic accuracy of integrating 2 imaging techniques—cardiac CT and ^{18}F-FDG PET/CT—in a relatively large cohort of patients with cardiac masses.

Our study highlights the ability of both cardiac CT and ^{18}F-FDG PET/CT to correctly discriminate between benign and malignant cardiac masses. Among patients undergoing cardiac CT, a diagnostic gray zone was observed, namely patients exhibiting 3 or 4 markers. In this context, the metabolic tumor indices measured by ^{18}F-FDG PET/CT may provide pivotal information leading to an accurate diagnosis, as shown by our results.

Our findings may improve the current clinical practice as the diagnostic work-up of cardiac masses cannot rely on specific algorithms. In fact, the initial echocardiographic evaluation in this setting is often inconclusive and second-level imaging techniques are usually required.

In this scenario, CMR is considered as the reference imaging tool, due to its excellent tissue characterization and high spatial resolution leading to an accurate discrimination of benign and malignant cardiac lesions (4,17-19). However, the execution of CMR is limited in claustrophobic patients, in those unable to perform the required breath-holds, and it is contraindicated in patients with non-CMR-compatible devices. Moreover, not all centers have the necessary expertise to correctly interpret CMR images (20).

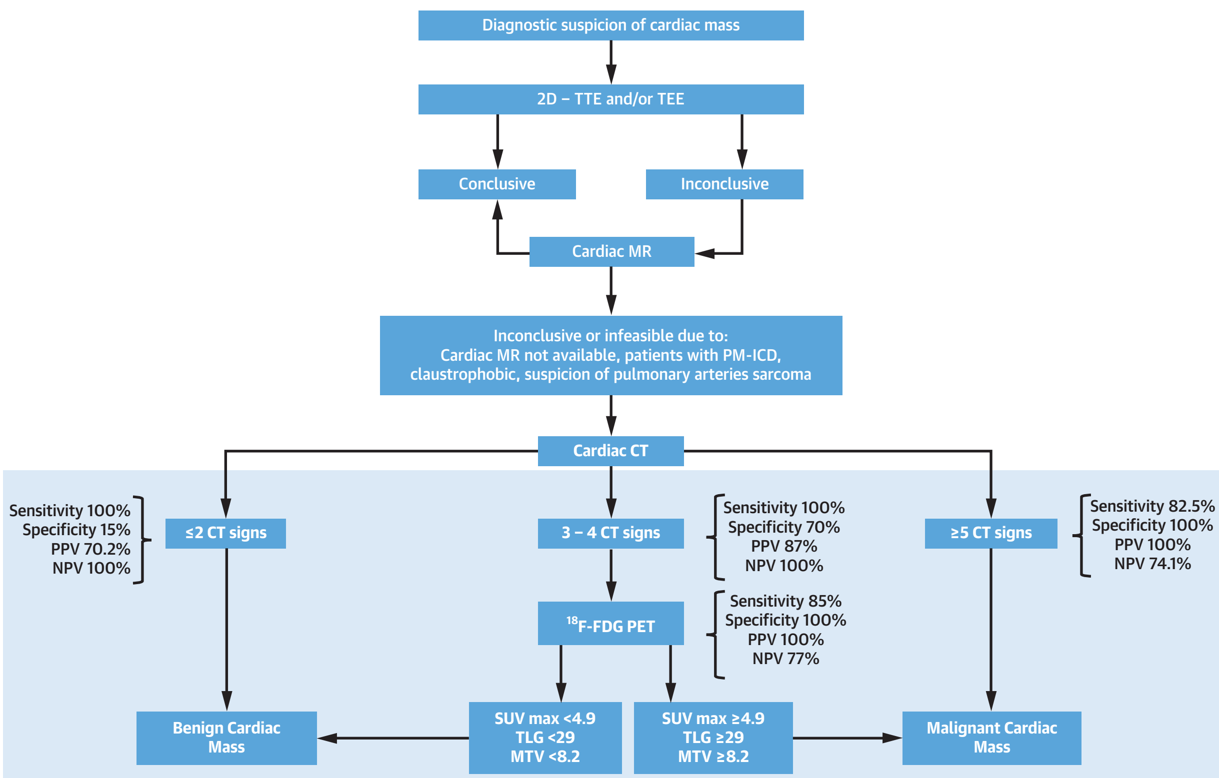
Cardiac CT has an emerging role as it is usually indicated in the suspicion of specific neoplasms (i.e., pulmonary arteries sarcoma) or when a disease staging is required (20,21).

Previous studies have analyzed the predictor role of various CT signs in identifying the nature of cardiac masses. Namely, the mass size and location and the enhancement pattern have been proposed as markers of malignancy (21,22).

In the present study, we assessed the diagnostic value of 8 pre-specified cardiac CT parameters to discriminate between benign and malignant cardiac masses. Notably, the diameter of cardiac masses, irregular cardiac mass margins, the presence of pericardial effusion, solid density, pre-contrast characteristics, and contrast uptake were found to be associated with malignancies. Calcifications did not differentiate the nature of the masses as the majority of pseudotumors exhibited this finding due to the underlying disease. Two malignant metastatic lesions showed CT calcifications, supposedly as a result of cellular necrosis and/or apoptosis or intra-tumoral hemorrhage, either primitive or chemotherapy-related (23).

On the other hand, our findings highlight the diagnostic role of contrast uptake also for benign tumors. In fact, as opposed to primary benign masses,

CENTRAL ILLUSTRATION Diagnostic Algorithm for Cardiac Masses



D'Angelo, E.C. et al. J Am Coll Cardiol Img. 2020;13(11):2400-11.

The algorithm shows a possible diagnostic approach to patients with suspected cardiac masses. **Bold** indicates the diagnostic role of cardiac computed tomography (CT) and ^{18}F-fluorodeoxyglucose (^{18}F-FDG) positron emission tomography (PET)/CT according to our findings. 2D = 2-dimensional; CMR = cardiac magnetic resonance; ICD = implantable cardioverter-defibrillator; MTV = metabolic tumor volume; NPV = negative predictive value; PM = pacemaker; PPV = positive predictive value; SUV = standardized uptake value; TEE = transesophageal echocardiogram; TLG = total lesion glycolysis; TTE = transthoracic echocardiogram.

only 1 of 12 pseudotumors appeared isodense after contrast enhancement CT. This was a case of pericardial cystic with an unusual enhancement pattern due to the chronically inflamed borders. The absent contrast uptake of pseudotumors is probably related to the avascularity of such lesions, as previously suggested (22,24).

Another novelty of our work is that we looked for a CT pattern able to identify the nature of cardiac masses. In particular, the presence of at least 5 CT signs was a perfect indicator of malignancy. This finding may have immediate clinical applications as CT is widely accessible and typically indicated in specific clinical scenarios such as disease-staging purposes or in cases with involvement of great vessels. Nevertheless, uncertainty exists in patients with 3 or 4 CT signs and another diagnostic tool is often required.

Only few limited series have evaluated the role of ^{18}F-FDG PET/CT in cardiac masses (11). These studies assessed ^{18}F-FDG quantification by determining SUV (11), the most commonly used semiquantitative parameter for tracer uptake analysis. Our study confirmed that quantification of ^{18}F-FDG uptake can help to distinguish benign from malignant masses. In fact, benign formations showed only a slight ^{18}F-FDG uptake, whereas in malignancies this index was very high (25,26). Furthermore, the availability of an SUV_{max} cutoff may be extremely useful to identify malignant lesions. In fact, none of the primary benign cardiac tumors had a SUV_{max} >4.9, whereas the majority of primary and secondary malignant tumors had a SUV_{max} higher than this cutoff. A previous study by Rahbar et al. (11) in 24 patients with cardiac tumors yielded 100% sensitivity and 86% specificity at a SUV_{max} cutoff of 3.5, whereas Nensa et al. (27)

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identified an optimal cutoff of 5.2 with a 100% sensitivity and 92% specificity for the differentiation between malignant and nonmalignant masses.

Another original feature of our study is that for the first time we reported the discriminant ability of MTV and TLG, indicators of tumor volume and activity, respectively (28-31). Existing data suggest that ^{18}F-FDG uptake in cancer cells is determined by several biologic variables such as number of tumor cells, tumor cell proliferation, and angiogenesis (11). Accordingly, our results showed that these indices were higher in cardiac malignant masses, whereas primary benign cardiac tumors had undetectable MTV and TLG.

Indeed, ^{18}F-FDG PET/CT is confirmed as an extremely powerful tool able to provide substantial information regarding the nature of cardiac masses. Specifically, our data highlight the utility of PET in the context of uncertain diagnoses after cardiac CT, namely patients with 3 or 4 abnormal CT signs. In such cases, the coexistence of all PET parameters below the cutoff values suggests a benign mass, whereas the detection of the least 1 abnormal PET feature perfectly identifies malignancies.

Undoubtedly, our work confirms the troubled diagnostic work-up of patients with cardiac masses. To simplify such a long and winding pathway, we propose a diagnostic algorithm potentially applicable in this setting (**Central Illustration**). Whereas CMR plays a central role following the initial echocardiographic evaluation, cardiac CT and ^{18}F-FDG PET/CT should be performed in specific clinical settings or for disease-staging purposes.

STUDY LIMITATIONS. First, our study population was recruited retrospectively and prospectively, although cardiac CT and ^{18}F-FDG PET/CT were performed with the same protocols.

Additionally, this study was conducted in a single institution with a relatively small number of patients, even though this is the largest series of histologically confirmed cardiac masses.

In our cohort, the prevalence of malignant cardiac masses was overestimated probably due to the nature of the study; in fact, we enrolled only patients with a diagnostic uncertainty at echocardiographic evaluation. Moreover, we cannot exclude a selection bias as 35 patients without histological examination were excluded. Due to the limited sample size, we were not able to build a classification tree analysis based on a chi-square sequential algorithm. However, future prospective studies with larger study populations may allow the construction of sequential diagnostic algorithms.

CONCLUSIONS

Cardiac CT and ^{18}F-FDG PET/CT provide valuable information for the identification of cardiac masses. The possibility of noninvasively addressing the lesions' nature is particularly relevant for therapeutic as well as prognostic purposes and should be routinely considered in this setting.

Although CMR clearly plays an essential role, cardiac CT and ^{18}F-FDG PET/CT might improve the diagnostic yield in specific conditions as proposed in our algorithm. Nevertheless, prospective studies with larger samples are required to confirm and validate our findings.

AUTHOR RELATIONSHIP WITH INDUSTRY

The authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: The use of CT and ^{18}F-FDG PET/CT has been tested in a heterogeneous group of patients with cardiac masses, all with histological certainty. Irregular tumor margins, pericardial effusion, invasion, solid nature, diameter, CT contrast uptake, and pre-contrast characteristics were strongly associated with the malignant nature of masses. The presence of at least 5 CT signs was a perfect indicator of malignancy. Notably, cases with 3 or 4 CT signs represent a gray area that may benefit from the use of ^{18}F-FDG PET to correctly identify the mass nature.

TRANSLATIONAL OUTLOOK: The diagnostic approach to suspected cardiac masses is often challenging and not homogeneous as various centers usually rely on local expertise. Additional and larger studies are needed to validate these findings and to create a sequential diagnostic algorithm in this scenario.

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KEY WORDS cardiac computed tomography, ^{18}F-fluorodeoxyglucose with positron emission tomography/computed tomography, primary cardiac benign tumors, primary malignant tumors, pseudotumors, secondary malignant tumors

APPENDIX For supplemental methods, a figure, and tables, please see the online version of this paper.